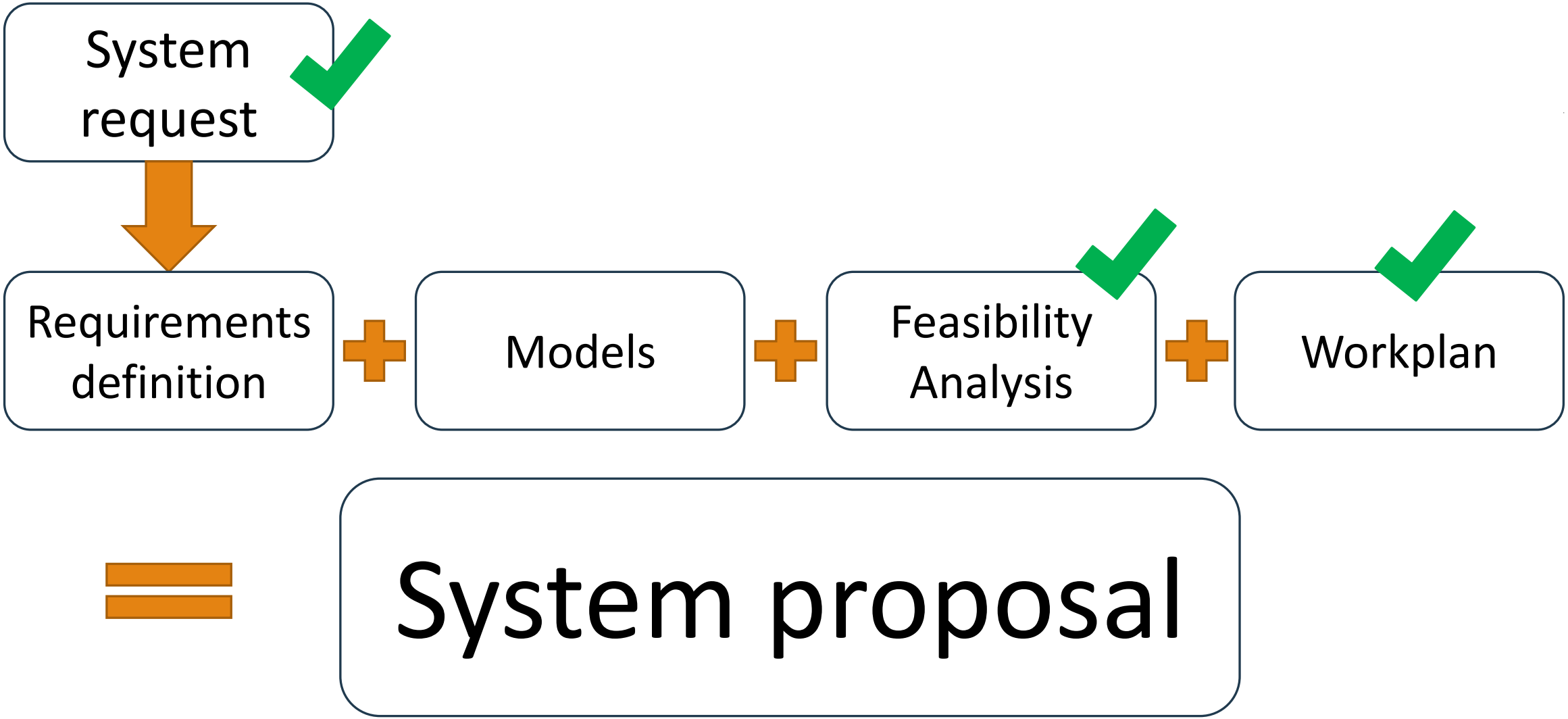
System Analysis

DR. DRAGUNOV

REQUIREMENTS DETERMINATION

Analysis phase



System proposal



Approval committee

????

The Difference Between Project and Product Requirements

Project Requirements

Project requirements are the high-level objectives, constraints, and conditions that need to be met for the project to be considered successful. They are usually defined by the project stakeholders and focus on the overall project's scope, timeline, budget, and resources.

Product Requirements

Product requirements, on the other hand, are the specific features, functionalities, and characteristics that the final product must possess to meet the needs of its end-users. They are typically defined by the product owner or the product management team and focus on the product's functionality, performance, and usability.

Project Requirements

Project requirements include:

Physical resources required by the development team, such as workspace, equipment, and so on.

Staff training needs.

Project documentation, including user manuals, and technical support documentation.

Infrastructure changes required for the software operating environment.

Requirements and procedures for installing, configuring, testing, and releasing the new product, as well as for migrating from the old system to the new one.

Product certification requirements, compliance with regulatory requirements, and legal protection (patents).

Internal company documents: policies, processes, structures, and so on.

Service Level Agreement (SLA) with customers. For example, warranty service.

Product Requirements, Requirements Definition Report or product requirements document

Summary: *A product requirements document (PRD) defines the requirements of a particular product, including the product's purpose, features, functionality, and behavior. It serves as a guide for business and technical teams to help build, launch, or market the product.*



Business requirements

One of the first activities of an analyst is to determine the business requirements for a new system.

We'll begin by presenting the requirements definition, a document that lists the new system's capabilities. It then describes how to analyze requirements using **business process automation**, **business process improvement**, and **business process reengineering** techniques and how to gather requirements using interviews, JAD sessions, questionnaires, document analysis, and observation. The chapter also describes a set of alternative requirements-documentation techniques and describes the system proposal document that pulls everything together.

OBJECTIVES

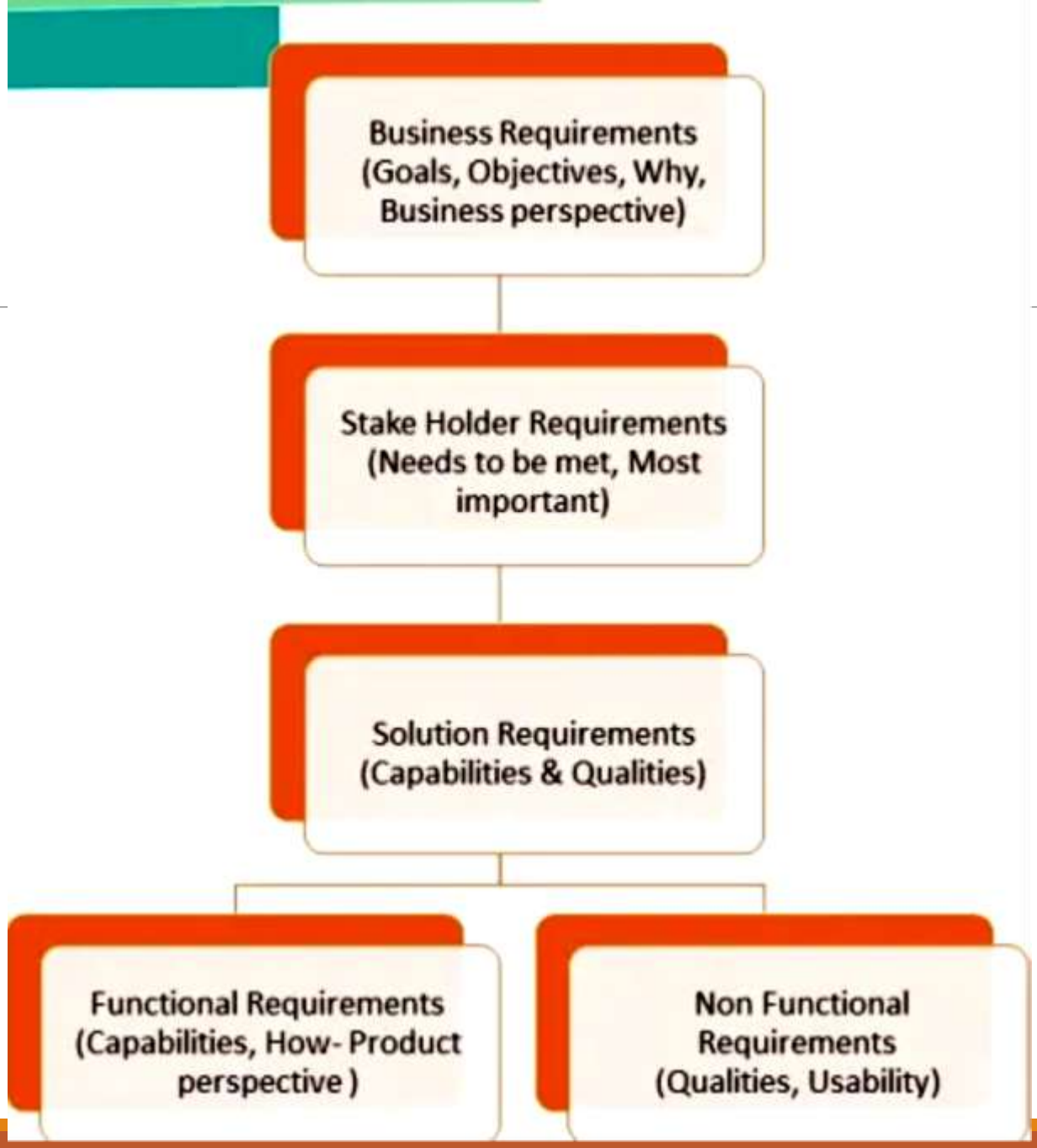
- Understand how to create a requirements definition
- Become familiar with requirements-analysis techniques
- Understand when to use each requirements-analysis technique
- Understand how to gather requirements using interviews, JAD sessions, questionnaires, document analysis, and observation
- Understand the use of concept maps, story cards, and task lists as requirements documentation techniques
- Understand when to use each requirements-gathering technique
- Be able to begin creating a system proposal

Defining a Requirement

During analysis, requirements are written from the perspective of the business person, and they focus on the "what" of the system.

Because they focus on the needs of the business user, they are usually called ***business requirements*** (and sometimes user requirements).

Later in design, business requirements evolve to *become more technical*, and they describe *how the system will be implemented*. Requirements in design are written from the developer's perspective, and they are usually called ***system requirements***.


Business Requirements (Goals, Objectives, Why, Business perspective)
Stake Holder Requirements (Needs to be met, Most important)
Solution Requirements (Capabilities & Qualities)
Functional Requirements (Capabilities, How- Product perspective)
Non Functional Requirements (Qualities, Usability)



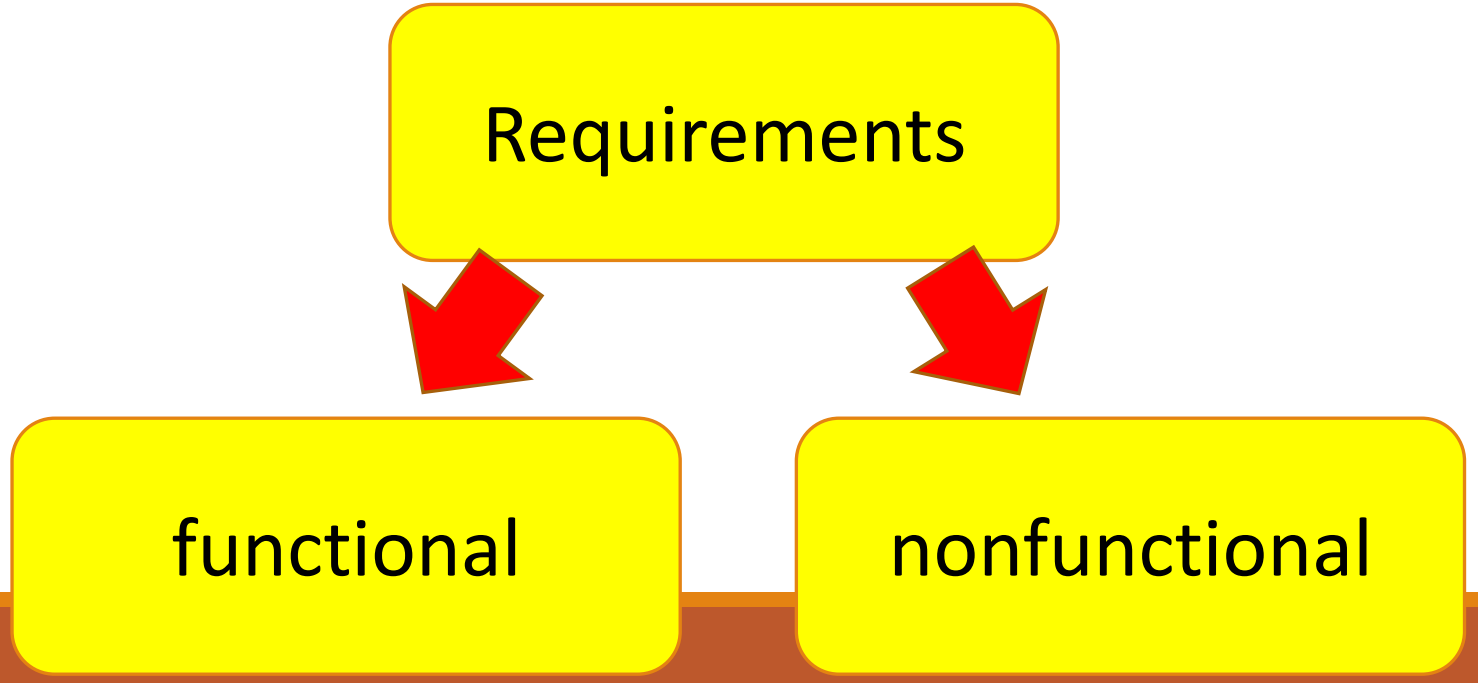
Purpose: to convert high level business requirements (from the system request) into detailed requirements that can be used as inputs for creating models

What is a requirement?

- **A statement of what the system must do or a characteristic it must have**
- Will later evolve into a technical description of how the system will be implemented

Types:

- **Functional**: relates to a process or data
- **Non-functional**: relates to performance or usability



Functional requirements

A functional requirement relates directly to a process a system has to perform or information it needs to contain.

Functional Requirement	Description	Examples
Process-oriented	A process the system must perform; a process the system must do	■ The system must allow registered customers to review their own order history for the past three years. ■ The system must check incoming customer orders for inventory availability. ■ The system should allow students to view a course schedule while registering for classes.
Information-oriented	Information the system must contain	■ The system must retain customer order history for three years. ■ The system must include real-time inventory levels at all warehouses. ■ The system must include budgeted and actual sales and expense amounts for current year and three previous years.

Nonfunctional requirements

Nonfunctional requirements refer to behavioral properties that the system must have, such as performance and usability. The ability to access the system using a Web browser is considered a nonfunctional requirement. Nonfunctional requirements can influence the rest of analysis (functional, structural, and behavioral models) but often do so only indirectly; nonfunctional requirements are used primarily in design when decisions are made about the user interface, the hardware and software, and the system's underlying physical architecture.

Nonfunctional Requirement	Description	Examples
Operational	The physical and technical environments in which the system will operate	■ The system should be able to fit in a pocket or purse. ■ The system should be able to integrate with the existing inventory system. ■ The system should be able to work on any Web browser.
Performance	The speed, capacity, and reliability of the system	■ Any interaction between the user and the system should not exceed 2 seconds. ■ The system should receive updated inventory information every 15 minutes. ■ The system should be available for use 24 hours per day, 365 days per year.
Security	Who has authorized access to the system under what circumstances	■ Only direct managers can see personnel records of staff. ■ Customers can see their order history only during business hours.
Cultural and political	Cultural, political factors and legal requirements that affect the system	■ The system should be able to distinguish between United States and European currency. ■ Company policy says that we buy computers only from Dell. ■ Country managers are permitted to authorize customer user interfaces within their units. ■ The system shall comply with insurance industry standards.

Source: The Atlantic Systems Guild, http://www.systemsguild.com/GuildSite/Robs/Template.html

One of the most common mistakes by new analysts is to confuse functional and nonfunctional requirements. Pretend that you received the following list of requirements for a sales system.

Requirements for Proposed System The system should:

1. be accessible to the Web users;
2. include the company standard logo and color scheme;
3. restrict access to profitability information;
4. include actual and budgeted cost information;
5. provide management reports;
6. include sales information that is updated at least daily;
7. have two-second maximum response time for predefined queries and ten-minute maximum response time for ad hoc queries;
8. include information from all company subsidiaries;
9. print subsidiary reports in the primary language of the subsidiary;
10. provide monthly rankings of salesperson performance

Requirements Definition Report

The requirements definition report—usually just called the requirements **definition—is a straightforward text report that simply lists the functional and nonfunctional requirements in an outline format.**

Next figure shows a sample requirements definition for an appointment system for a typical doctor's office.

Nonfunctional Requirements

1. Operational Requirements
- 1.1. The system will operate in Windows environment.
- 1.2. The system should be able to connect to printers wirelessly.
- 1.3. The system should automatically back up at the end of each day.

2. Performance Requirements
- 2.1. The system will store a new appointment in 2 seconds or less.
- 2.2. The system will retrieve the daily appointment schedule in 2 seconds or less.

3. Security Requirements
- 3.1. Only doctors can set their availability.
- 3.2. Only a manager can produce a schedule.

4. Cultural and Political Requirements
- 4.1. No special cultural and political requirements are anticipated.

Functional Requirements

1. Manage Appointments
- 1.1. Patient makes new appointment.
- 1.2. Patient changes appointment.
- 1.3. Patient cancels appointment.

2. Produce Schedule
- 2.1. Office Manager checks daily schedule.
- 2.2. Office Manager prints daily schedule.

3. Record Doctor Availability
- 3.1. Doctor updates schedule

Determining Requirements (strategies)

The most effective approach is to have both business people and analysts working together to determine business requirements. Sometimes, however, users don't know exactly what they want, and analysts need to help them discover their needs.

Three kinds of strategies have become popular to help analysts do this:

- Business process automation (BPA)
- Business process improvement (BPI)
- Business process reengineering (BPR).

Analysts can use these tools when they need to guide the users in explaining what is wanted from a system.

The three kinds of strategies work similarly. They help users critically examine the current state of systems and processes (the as-is system), identify exactly what needs to change, and develop a concept for a new system (the to-be system). A different amount of change is associated with each technique;

- BPA creates a small amount of change
- BPI creates a moderate amount of change
- BPR creates significant change that affects much of the organization.

Determining Requirements (techniques)

Therefore, analysts use a portfolio of requirements-gathering techniques to acquire information from users. The analyst has many techniques from which to choose:

- interviews,
- questionnaires,
- observation,
- joint application development (JAD)
- document analysis.

The information gathered using these techniques is critically analyzed and used to craft the requirements definition report.

Real-World Problems with Requirements Determination

- First, the analyst might not have access to the correct set of users to uncover the complete set of requirements.
- Second, the specification of the requirements may be inadequate. This can be especially true with the lightweight techniques associated with agile methodologies.
- Third, some requirements are simply unknowable at the beginning of a development process.
- Fourth, verifying and validating of requirements can be very difficult.

REQUIREMENTS ANALYSIS STRATEGIES

Before the project team can determine what requirements are appropriate for a given system, they need to have a clear vision of the kind of system that will be created and the level of change that it will bring to the organization. The basic process of analysis is divided into three steps:

- understanding the as-is system
- identifying improvements
- developing requirements for the to-be system.



Business Process Automation (BPA)

BPA leaves the basic way the organization operates unchanged and uses computer technology to do some of the work. **BPA can make the organization more efficient but has the least impact on the business.** Planners in BPA projects spend a significant time understanding the current as-is system before moving on to improvements and to-be system requirements. Problem analysis and root cause analysis are two popular BPA techniques.

- **Problem Analysis**
- **Root Cause Analysis**

- **Problem Analysis** The most straightforward (and probably the most commonly used) requirements-analysis technique is *problem analysis*. Problem analysis means asking the users and managers to identify problems with the as-is system and to describe how to solve them in the to-be system.

- **Root Cause Analysis** The ideas produced by problem analysis tend to be solutions to problems. **All solutions make assumptions about the nature of the problem, assumptions that might or might not be valid.** In our experience, users (and most people in general) tend to quickly jump to solutions without fully considering the nature of the problem.

  Sometimes the solutions are appropriate, but many times they address a *symptom* of the problem, not the true problem or *root cause* itself.

Business Process Improvement (BPI)

BPI makes moderate changes to the way the organization operates in order to take advantage of new opportunities offered by technology or to copy what competitors are doing.

BPI can improve efficiency (i.e., doing things right) and improve effectiveness (i.e., doing the right things). Planners of BPI projects also spend time understanding the as-is system, but much less time than with BPA projects; their primary focus is on improving business processes, so time is spent on the as-is only to help with the improvement analyses and the to-be system requirements.

Popular BPI activities

- Duration analysis
- activity-based costing
- informal benchmarking

Duration Analysis

Duration analysis requires a detailed examination of the amount of time it takes to perform each process in the current as-is system. The analysts begin by determining the total amount of time it takes, on average, to perform a set of business processes for a typical input. They then time each of the individual steps (or subprocesses) in the business process. The time to complete the basic steps are then totaled and compared to the total for the overall process.

Example

For example, suppose that the analysts are working on a home mortgage system and discover that on average, it takes thirty days for the bank to approve a mortgage. They then look at each of the basic steps in the process (e.g., data entry, credit check, title search, appraisal) and find that the total amount of time actually spent on each mortgage is about eight hours. This is a strong indication that the overall process is badly broken, because it takes thirty days to perform one day's work.

Activity-Based Costing

Activity-based costing is a similar analysis; it examines the cost of each major process or step in a business process rather than the time taken. The analysts identify the costs associated with each of the basic functional steps or processes, identify the most costly processes, and focus their improvement efforts on them.

Assigning costs is conceptually simple. Analysts simply examine the direct cost of labor and materials for each input. Materials costs are easily assigned in a manufacturing process, whereas labor costs are usually calculated based on the amount of time spent on the input and the hourly cost of the staff. However, as you may recall from a managerial accounting course, there are indirect costs such as rent, depreciation, and so on, that also can be included in activity costs.

Informal Benchmarking

Benchmarking refers to studying how other organizations perform a business process in order to learn how your organization can do something better.

Benchmarking helps the organization by introducing ideas that employees may never have considered but that have the potential to add value. Informal benchmarking is fairly common for customer-facing business processes (i.e., processes that interact with the customer).

With informal benchmarking, the managers and analysts think about other organizations or visit them as customers to watch how the business process is performed. In many cases, the business studied may be a known leader in the industry or simply a related firm. For example, suppose the team is developing a website for a car dealer. The project sponsor, key managers, and key team members would likely visit the websites of competitors as well as those of others in the car industry (e.g., manufacturers, accessories suppliers) and those in other industries that have won awards for their websites.

Business Process Reengineering

BPR means changing the fundamental way the organization operates, obliterating the current way of doing business and making major changes to take advantage of new ideas and new technology. Planners of BPR projects spend little time understanding the as-is, because their goal is to focus on new ideas and new ways of doing business.

Popular BPR activities

- Outcome analysis
- Technology analysis
- Activity elimination

Outcome Analysis

Outcome analysis focuses on understanding the fundamental outcomes that provide value to customers. Although these outcomes sound as though they should be obvious, they often are not.

Technology Analysis

Many major changes in business since the turn of the century have been enabled by new technologies. *Technology analysis* starts by having the analysts and managers develop a list of important and interesting technologies. Then the group systematically identifies how every technology could be applied to the business process and identifies how the business would benefit.

Activity Elimination

Activity elimination is exactly what it sounds like. The analysts and managers work together to identify how the organization could eliminate each activity in the business process, how the function could operate without it, and what effects are likely to occur. Initially, managers are reluctant to conclude that processes can be eliminated, but this is a force-fit exercise in that they must eliminate each activity. In some cases the results are silly; nonetheless, participants must address every activity in the business process.

Selecting Appropriate Strategies

Each technique discussed has its own strengths and weaknesses. No one technique is inherently better than the others, and in practice most projects use a combination of techniques.

	Business Process Automation	Business Process Improvement	Business Process Reengineering
Potential business value	Low–moderate	Moderate	High
Project cost	Low	Low–moderate	High
Breadth of analysis	Narrow	Narrow–moderate	Very broad
Risk	Low–moderate	Low–moderate	Very high

REQUIREMENTS-GATHERING TECHNIQUES

The requirements-gathering process is used for building political support for the project and establishing trust and rapport between the project team building the system and the users who ultimately will choose to use or not use the system. Involving someone in the process implies that the project teams view that person as an important resource and value his or her opinions.

All the key stakeholders must be included in the requirements-gathering process.

The second challenge of requirements gathering is choosing the way(s) information is collected. There are many techniques for gathering requirements that vary from asking people questions to watching them work.

Interviews

An interview is the most commonly used requirements-gathering technique. After all, it is natural— if you need to know something, you usually ask someone. In general, interviews are conducted one-on-one (one interviewer and one interviewee), but sometimes, owing to time constraints, several people are interviewed at the same time. There are five basic steps to the interview process:

- selecting interviewees
- designing interview questions
- preparing for the interview
- conducting the interview
- postinterview follow-up.

The first step in interviewing is to create an *interview schedule* listing all the people who will be interviewed, when, and for what purpose

Name	Position	Purpose of Interview	Meeting
Andria McClellan	Director, Accounting	Strategic vision for new accounting system	Mon., March 1 8:00–10:00 AM
Jennifer Draper	Manager, Accounts Receivable	Current problems with accounts receivable process; future goals	Mon., March 1 2:00–3:15 PM
Mark Goodin	Manager, Accounts Payable	Current problems with accounts payable process; future goals	Mon., March 1 4:00–5:15 PM
Anne Asher	Supervisor, Data Entry	Accounts receivable and payable processes	Wed., March 3 10:00–11:00 AM
Fernando Merce	Data Entry Clerk	Accounts receivable and payable processes	Wed., March 3 1:00–3:00 PM

People at different levels of the organization have different perspectives on the system, so it is important to include both managers who manage the processes and staff who actually perform the processes to gain both high-level and low-level perspectives on an issue.

There are three types of interview questions: **closed-ended questions**, **open-ended questions**, and **probing questions**.

Closed-ended questions are those that require a specific answer. They are similar to multiple-choice or arithmetic questions on an exam. Closed-ended questions are used when an analyst is looking for specific, precise information (e.g., how many credit card requests are received per day).

Types of Questions	Examples
Closed-ended questions	• How many telephone orders are received per day? • How do customers place orders? • What information is missing from the monthly sales report?
Open-ended questions	• What do you think about the current system? • What are some of the problems you face on a daily basis? • What are some of the improvements you would like to see in a new system?
Probing questions	• Why? • Can you give me an example? • Can you explain that in a bit more detail?

Open-ended questions are those that leave room for elaboration on the part of the interviewee. They are similar in many ways to essay questions that you might find on an exam. Open-ended questions are designed to gather rich information and give the interviewee more control over the information that is revealed during the interview.

The third type of question is the ***probing question***. Probing questions follow up on what has just been discussed in order to learn more, and they often are used when the interviewer is unclear about an interviewee's answer. They encourage the interviewee to expand on or to confirm information from a previous response, and they signal that the interviewer is listening and is interested in the topic under discussion.

No type of question is better than another, and a combination of questions is usually used during an interview. At the initial stage of an IS development project, the as-is process can be unclear, so the interview process begins with ***unstructured interviews,*** interviews that seek broad and roughly defined information. In this case, the interviewer has a general sense of the information needed but has few closed-ended questions to ask. These are the most challenging interviews to conduct because they require the interviewer to ask open-ended questions and probe for important information on the fly.

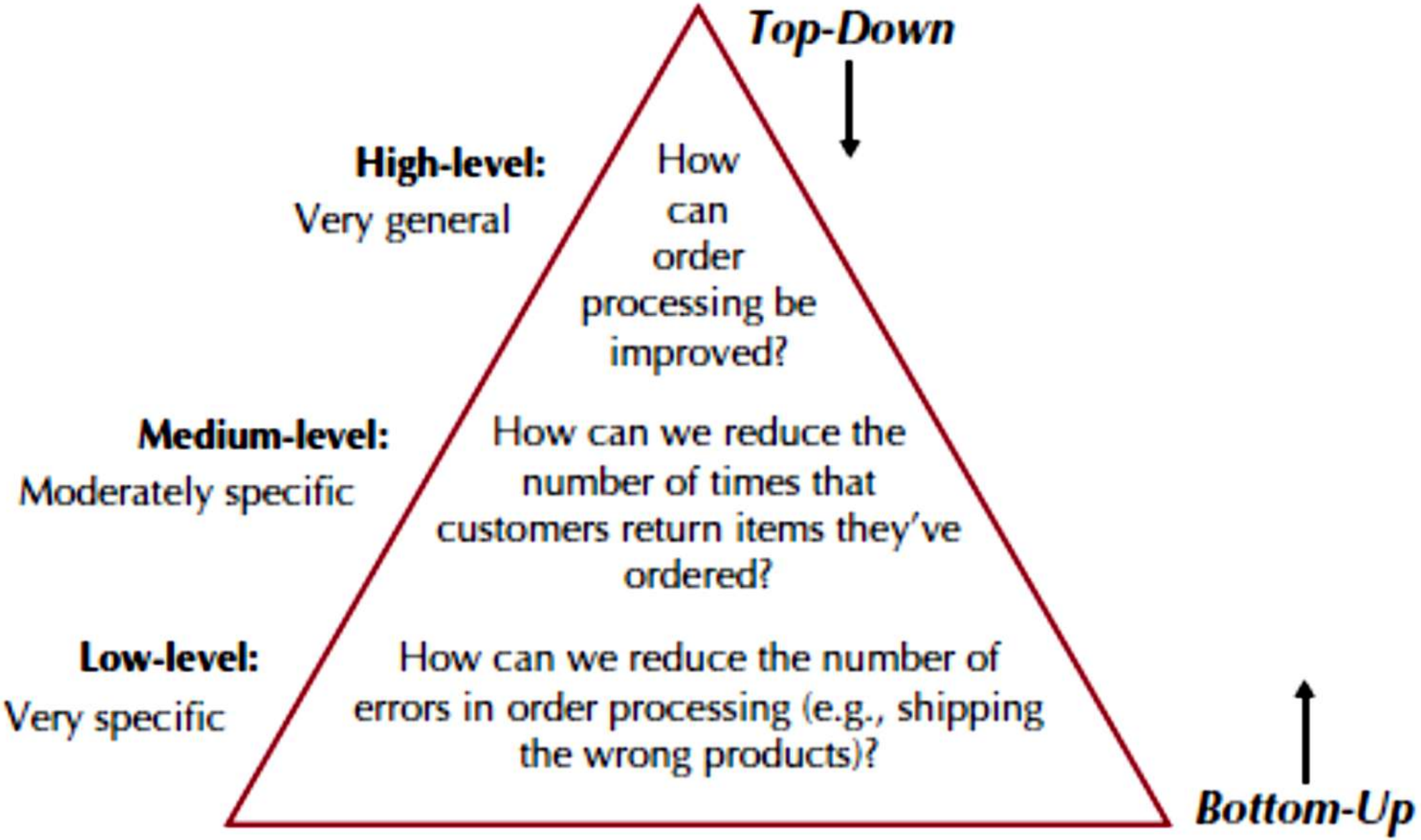
As the project progresses, the analyst comes to understand the business process much better and needs very specific information about how business processes are performed (e.g., exactly how a customer credit card is approved). At this time, the analyst conducts ***structured interviews,*** in which specific sets of questions are developed before the interviews. There usually are more closed-ended questions in a structured interview then in the unstructured approach.

No matter what kind of interview is being conducted, interview questions must be organized into a logical sequence so that the interview flows well.

There are two fundamental approaches to organizing the interview questions: **top down or bottom up.**

With the ***top-down interview***, the interviewer starts with broad, general issues and gradually works toward more-specific ones.

With the ***bottom-up interview***, the interviewer starts with very specific questions and moves to broad questions. In practice, analysts mix the two approaches, starting with broad, general issues, moving to specific questions, and then returning to general issues.


Top-Down
High-level:
Very general
How can order processing be improved?
Medium-level:
Moderately specific
How can we reduce the number of times that customers return items they've ordered?
Low-level:
Very specific
How can we reduce the number of errors in order processing (e.g., shipping the wrong products)?
Bottom-Up

Prepare for the Interview

It is important to prepare for the interview in the same way that you would prepare to give a presentation. The interviewer should have a general interview plan listing the questions to be asked in the appropriate order, should anticipate possible answers and provide follow-up with them, and should identify segues between related topics. The interviewer should confirm the areas in which the interviewee has knowledge so as not to ask questions that the interviewee cannot answer. Review the topic areas, the questions, and the interview plan, and clearly decide which have the greatest priority in case time runs short.

Conduct the Interview

In starting the interview, the first goal is to build rapport with the interviewee, so that he or she trusts the interviewer and is willing to tell the whole truth, not just give the answers that he or she thinks are wanted. The interviewer should appear to be professional and an unbiased, independent seeker of information. The interview should start with an explanation of why the interviewer is there and why he or she has chosen to interview the person; then the interviewer should move into the planned interview questions.

It is critical to carefully record all the information that the interviewee provides. In our experience, the best approach is to take careful notes—write down everything the interviewee says, even if it does not appear immediately relevant. The interviewer shouldn't be afraid to ask the person to slow down or to pause while writing, because this is a clear indication that the interviewee's information is important.

Post-Interview Follow-up

After the interview is over, the analyst needs to prepare an interview report that describes the information from the interview. The report contains interview notes, information that was collected over the course of the interview and is summarized in a useful format. In general, the interview report should be written within forty-eight hours of the interview, because the longer the interviewer waits, the more likely he or she is to forget information.

Often, the interview report is sent to the interviewee with a request to read it and inform the analyst of clarifications or updates.

Interview Notes Approved by: Linda Estey

Person Interviewed: Linda Estey,
Director, Human Resources

Interviewer: Barbara Wixom

Purpose of Interview:

- Understand reports produced for Human Resources by the current system
- Determine information requirements for future system

Summary of Interview:

- Sample reports of all current HR reports are attached to this report. The information that is not used and missing information are noted on the reports.
- Two biggest problems with the current system are:
  1. The data are too old (the HR Department needs information within two days of month end; currently information is provided to them after a three-week delay)
  2. The data are of poor quality (often reports must be reconciled with departmental HR database)
- The most common data errors found in the current system include incorrect job level information and missing salary information.

Open Items:

- Get current employee roster report from Mary Skudrna (extension 4355).
- Verify calculations used to determine vacation time with Mary Skudrna.
- Schedule interview with Jim Wack (extension 2337) regarding the reasons for data quality problems.

Detailed Notes: See attached transcript.